MoonEP

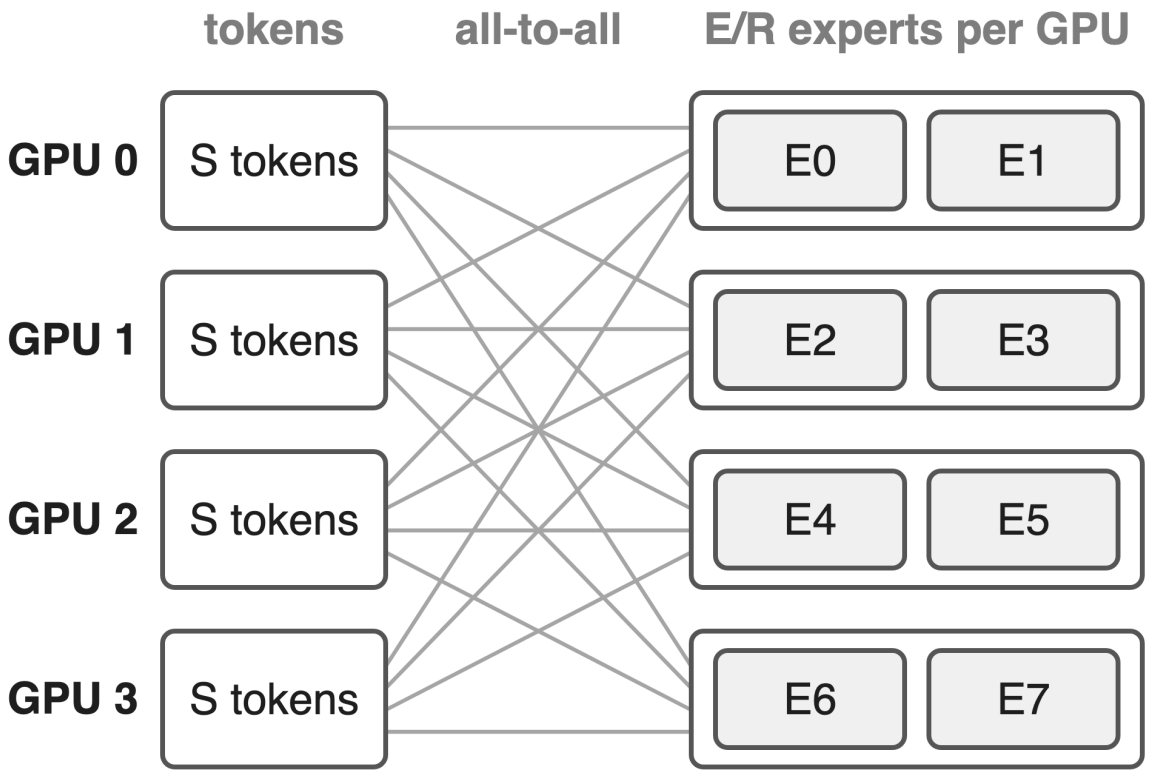
Perfectly balanced expert-parallel communication library

Moonshot AI · Kimi K3

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Kimi K3 trains a 2.8T MoE with expert parallelism

- **896** routed experts, **16** per token
- Each GPU hosts E/R experts
- Two all-to-alls per MoE layer: dispatch, combine



Skewed routing crashes DeepEP v2 training

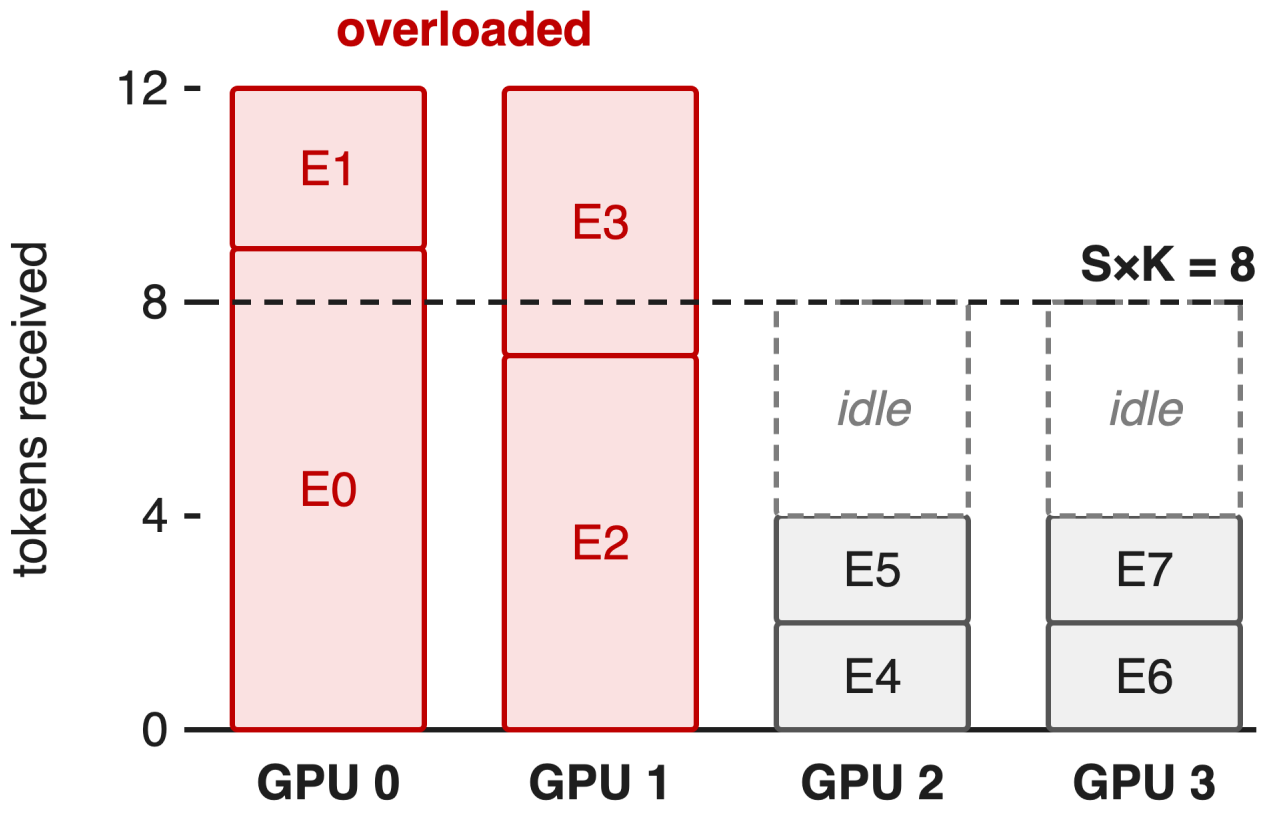
OOM

DeepEP v2 end-to-end training at maxvio 20, where the hottest expert carries 21× the average load

MoonEP README, e2e benchmark: 8×H20, EP=8, 9 layers, H=7168, E=256, top-8, seq 16k

The hottest GPU sets every layer's time

- Example: 4 GPUs, each sends S×K = 8 copies
- GPU 0-1 receive 12, GPU 2-3 only 4
- Counts change per micro-batch: **host sync** or **S×K×R buffers**



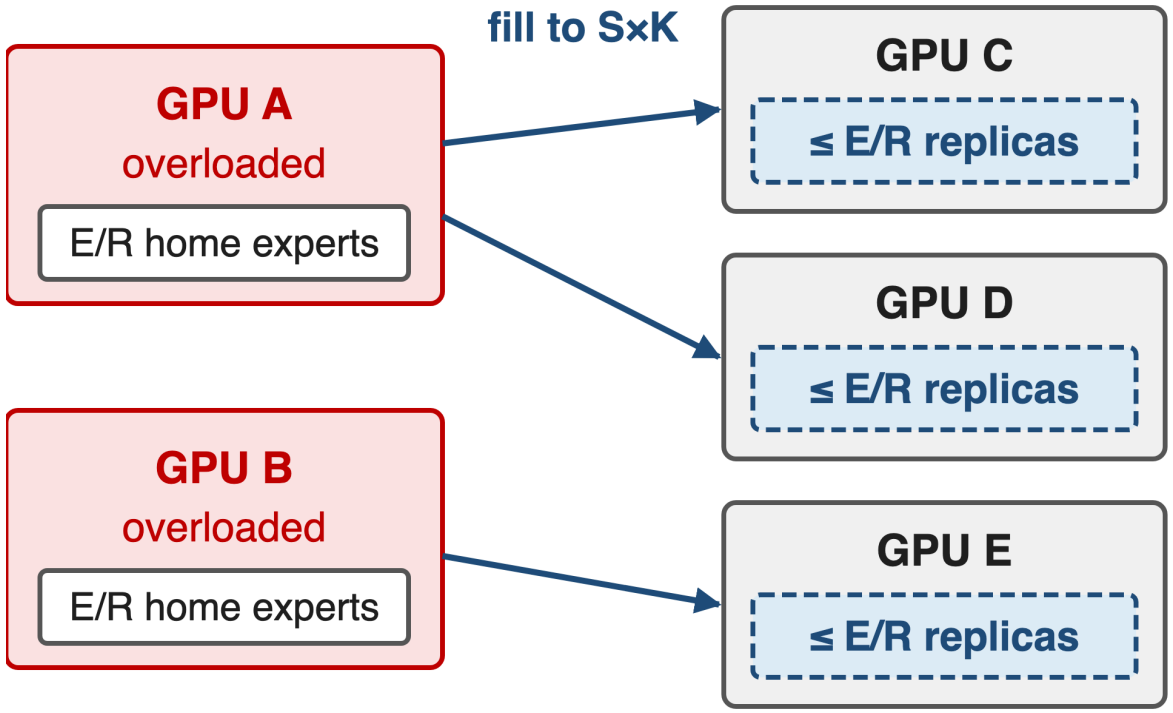
Balancing on average misses each micro-batch

approach	balances	updated	failure mode
Router bias (decks 55, 56)	batch-average expert load	from the previous step	**one micro-batch can still skew**
Offline replicas (EPLB, DeepSeek-V3)	historical GPU load	~every 10 minutes	**stale when routing drifts**
Capped cloning (ECHO, UltraEP)	current micro-batch	every forward	**stops when no plan fits the cap**

Needed: exact balance for this micro-batch, always feasible, planned on the GPU

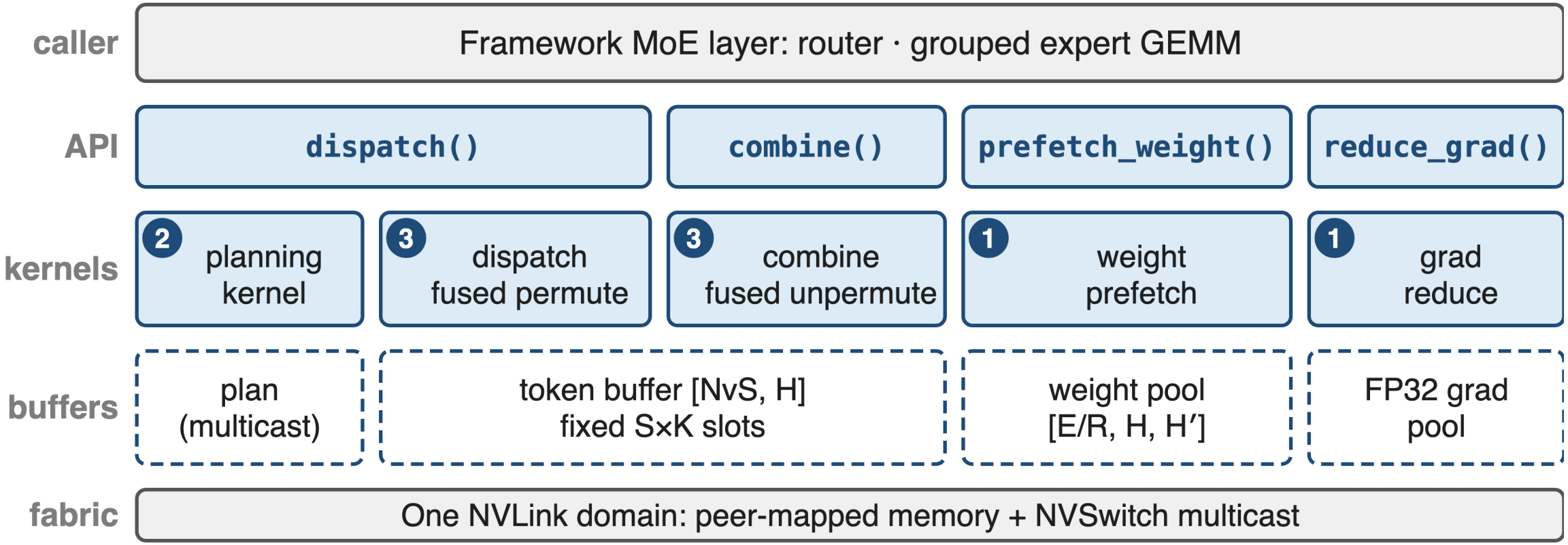
Insight: E/R replica slots always suffice

- Fill each under-loaded GPU from **one** overloaded GPU
- Its replicas come from E/R home experts
- Tight: some routings need ≈ E/R



Every GPU always gets exactly S×K tokens, so every shape is static

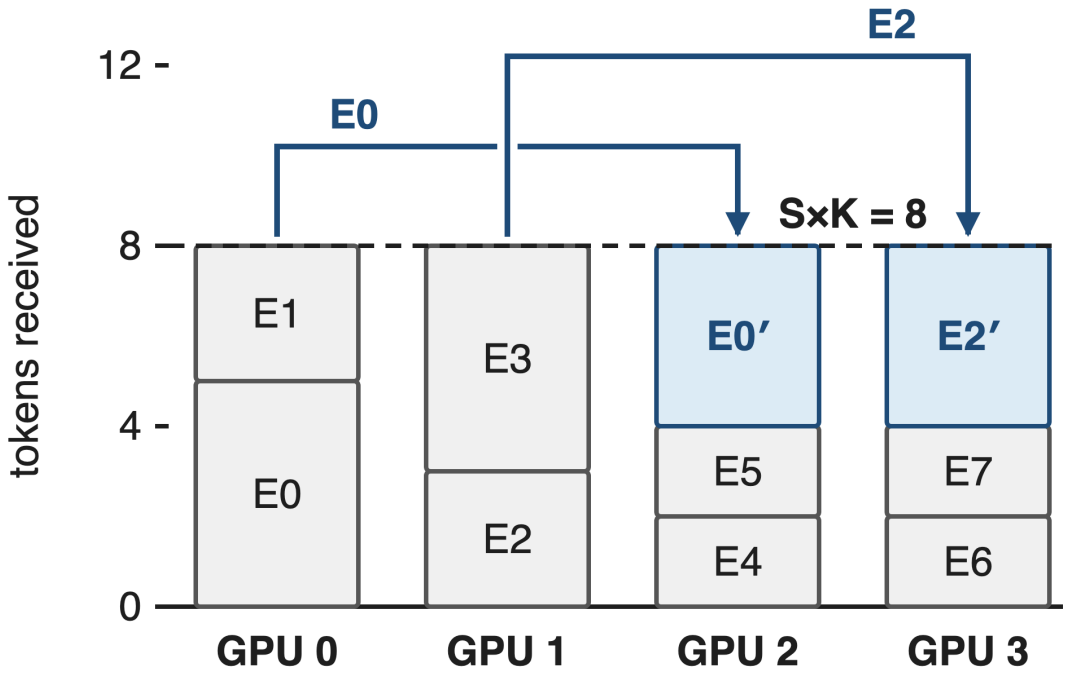
MoonEP stacks four calls on NVLink buffers



❶ dynamic redundant experts · ❷ online GPU planning · ❸ zero-copy static shapes (deep dive: deck 20)

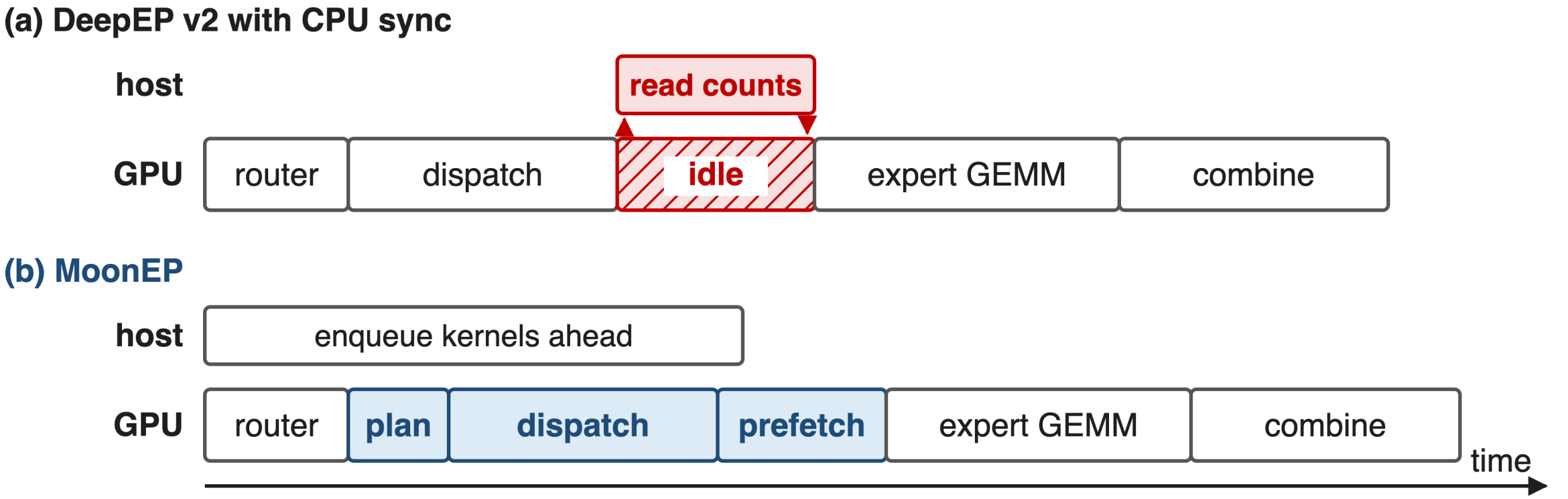
❶ Hot experts get replicas for one micro-batch

- Plan sends part of E0, E2 to GPU 2-3
- Forward: **prefetch** replica weights
- Backward: **reduce** replica grads home



Every GPU computes exactly S×K tokens; gradients match the unreplicated model

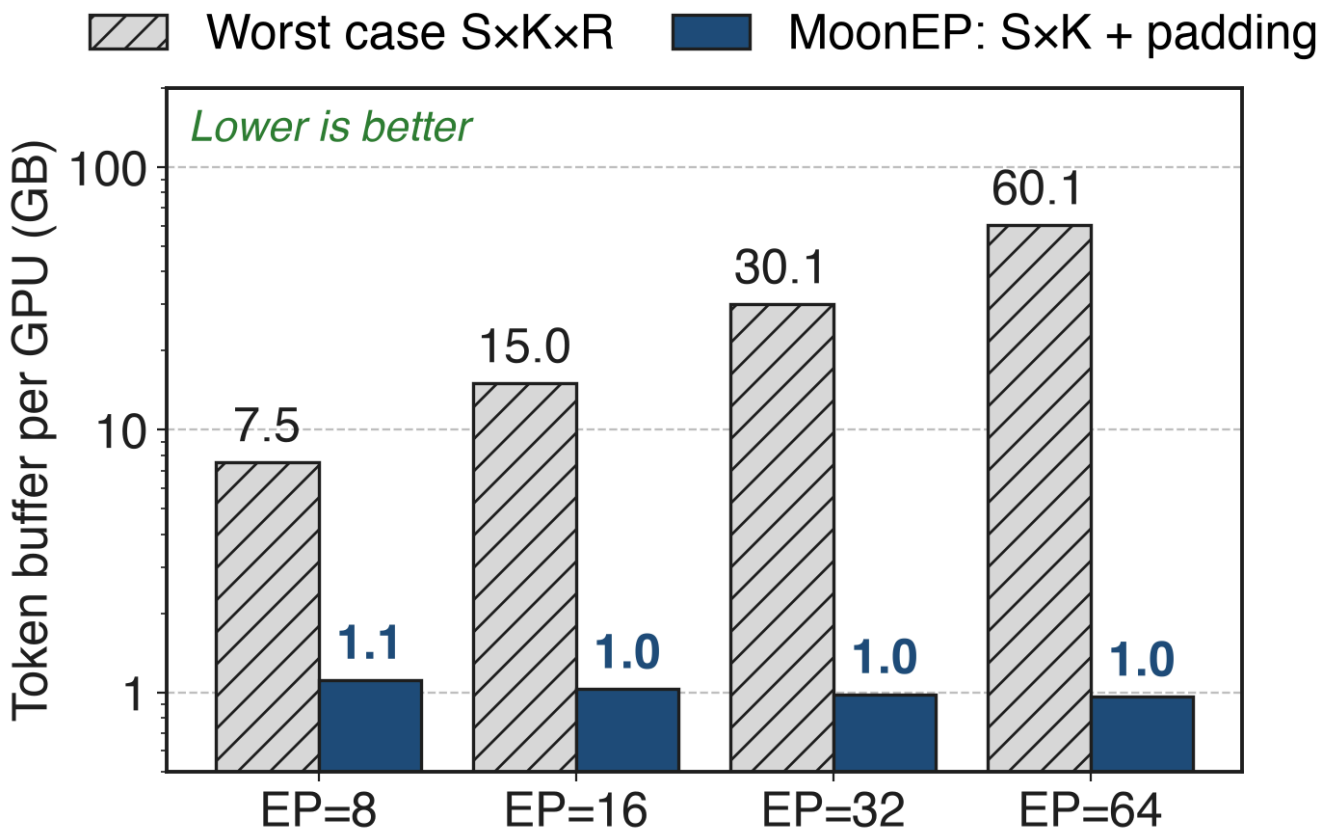
❷ One GPU kernel plans every layer



Planning ≈ 2% of dispatch time (≈ 45 µs), per micro-batch and layer

❸ Balance pins the token buffer at S×K

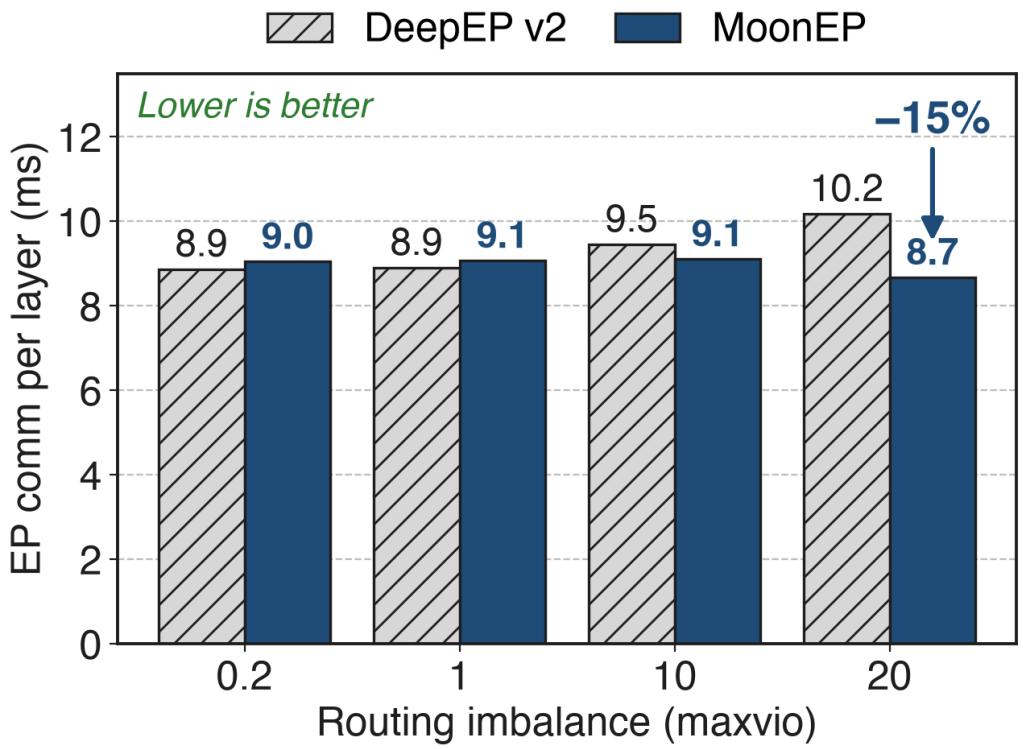
- Tokens land directly in expert-grouped slots
- GEMM reads buffer views: **no copies**
- Fixed shapes: **no per-layer host sync**



Analytic: BF16, S=8192, K=8, H=7168, E=384

EP comm stays flat; DeepEP v2 grows 15%

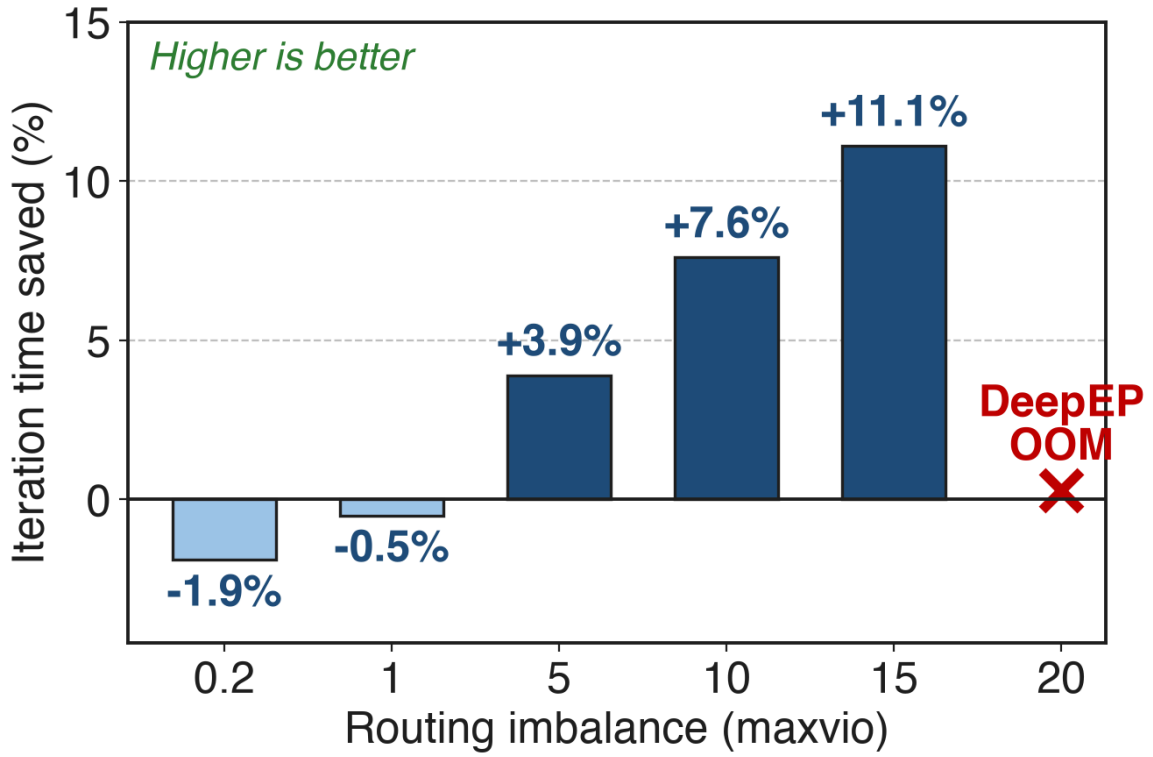
- Dispatch + combine, forward + backward
- MoonEP bars include plan + prefetch
- 8×H20, EP=8, E=384, S=8192



+2% when balanced (maxvio 0.2), –15% at maxvio 20

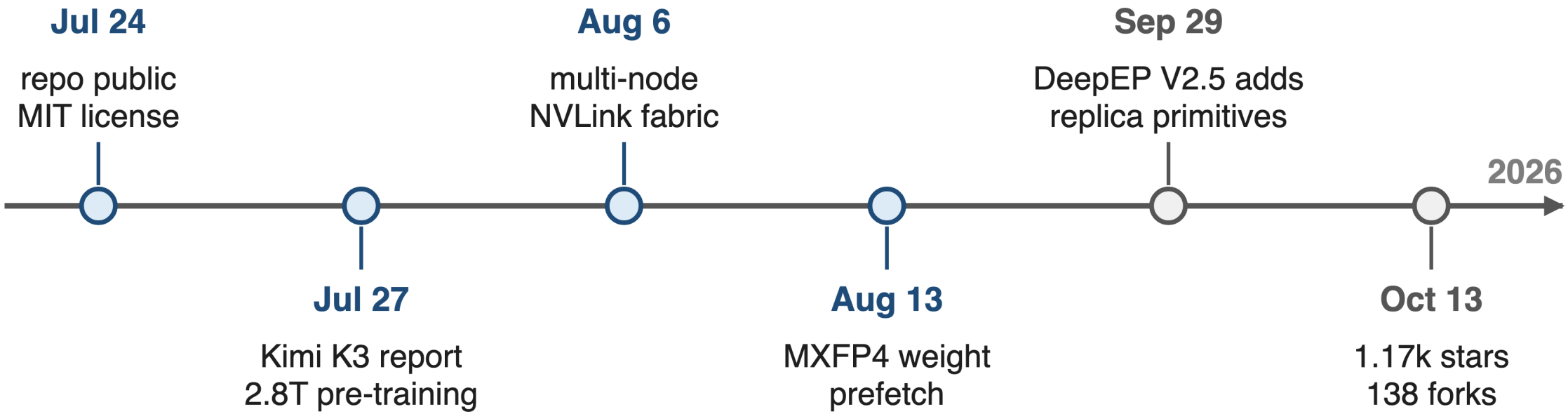
Iterations up to 11.1% faster; DeepEP OOMs at 20

- 9-layer model: E=256, top-8, seq 16k
- MoonEP iteration time flat across maxvio
- Static shapes: no fragmentation, no OOM



Balance has a small price: –1.9% at maxvio 0.2

Kimi K3 runs on it; DeepEP added its primitives



K3's 2.8T pre-training runs on MoonEP; DeepEP V2.5 ships its replica prefetch and grad reduce

Costs: prefetch time, a weight pool, NVLink only

- **Prefetch**: ≈0.5-0.75 ms per dispatch
- **Pool**: E/R experts + FP32 grads: 4.2 + 8.5 GB at EP=8
- **Scope**: one NVLink domain; per-expert skew remains

Pays off under skew; near balance, DeepEP v2 is slightly faster

Summary

- Skew overloads the hottest GPU; DeepEP v2 OOMs at maxvio 20
- **Insight**: ≤ E/R replicas per GPU give exactly S×K tokens
- ❶ per-micro-batch replicas · ❷ GPU planning · ❸ zero-copy buffer
- **–15%** EP comm, **11.1%** faster steps under skew
- **1.1 GB** vs **7.5 GB** token buffer at EP=8

https://github.com/MoonshotAI/MoonEP